

4(a)	any two points from: • signal can be regenerated/noise can be removed • signal can be encrypted • signal can be checked for errors • multiplexing is possible • <u>circuits</u> are more reliable/cheaper • <u>data</u> can be transmitted at a greater <u>rate</u>	B2
4(b)(i)	right-hand zero underlined (011 <u>0</u>)	B1
4(b)(ii)	analogue signals given as: 3.0, 4.8, 1.0	B1
	0011 at 0.30 ms and 0001 at 0.50 ms	B1
	0100 at 0.40 ms	B1
4(c)	series of steps, all of width 0.1 ms	B1
	steps levels, in order, at output voltage 0, 5, 6, 3 and 4 mV 2 marks: all levels correct 1 mark: one level incorrect and all others correct or one level omitted and last step shown at 1 mV	B2